

Confidential Legal Translation — Offline Workflow Architecture

Prepared 2026-08-01 · EN ↔ SL, later DE · Target machine: hpelitebook8g1i16

1. Headline finding

Slovene has a purpose-built open-weight model, developed at the University of Ljubljana specifically because general models underserve the language. Its availability resolves the model-selection question and displaces the earlier provisional recommendation.

It also relocates the project risk. Model choice is now the easiest decision in this workflow. The three genuinely difficult problems are optical character recognition of scanned evidence, terminology consistency across roughly one hundred documents, and format-preserving round-trip of the spreadsheet. Each of these will affect the certifying translator's editing burden more than the choice between one language model and another.

Scope note. This document addresses technical architecture only. Section 9 raises professional and procedural questions that fall outside it and should be resolved with the certifying translator before technical work begins, because a negative answer there would invalidate the entire approach.

2. Revised model selection

2.1 Primary translator: GaMS3-12B-Instruct

GaMS (Generative Model for Slovene) is developed by CJVT at the University of Ljubljana under the PoVeJMo research program. The current generation adapts Google's Gemma 3 through three-stage continual pre-training on approximately 140 billion tokens of Slovene, English, Bosnian, Serbian, and Croatian text, followed by two-stage supervised fine-tuning on over 200,000 examples.

Two published results make it the correct default here. It outperforms the base Gemma 3 12B model across Slovene benchmarks, English-to-Slovene translation, and the Slovene LLM arena; and in that arena it performs comparably to the much larger commercial GPT-4o, with a win rate above 60 percent. English-to-Slovene translation was an explicit evaluation target rather than an incidental capability.

The scale of the underlying problem explains why a dedicated model matters so much. Available authentic Slovene corpora total roughly 40 billion tokens, against the 30 trillion tokens used to train a contemporary frontier model — a ratio of about one to a thousand. General multilingual models are not merely weaker on Slovene; they are trained on a fundamentally different order of data.

2.2 Quantization: choose quality over speed

Quantization degrades low-resource language performance disproportionately. The linguistic patterns that a model has seen least often are the first to be lost when precision is reduced, and Slovene morphology — six cases, dual number, aspectual pairs — is exactly the kind of fine structure that suffers. Because this is an unattended batch workload rather than an interactive one, throughput should be traded away for accuracy.

Quant	File size	Est. speed	Assessment
Q6_K	9.66 GB	~6–7 tok/s	Recommended. Near-lossless in practice, comfortable memory headroom for long context
Q8_0	~13 GB	~5 tok/s	Acceptable if a measured quality difference against Q6_K justifies the throughput cost
Q5_K_M	8.45 GB	~7–8 tok/s	Reasonable fallback if Q6_K proves too slow in practice

Quant	File size	Est. speed	Assessment
Q4_K_M	7.30 GB	~8–9 tok/s	Not recommended for certified legal output at this language pair

Available memory is 27 GiB, so even Q8_0 leaves ample room. There is no hardware reason to economize here.

2.3 Supporting models

Model	Role	Rationale
qwen3.6:35b-a3b	English-side utility	Already installed and roughly four times faster. Use for glossary extraction, consistency checking of English output, and batch orchestration — not for Slovene generation
Gemma 3 / 4 12B	German pairs, later phase	German is high-resource and well served by the base Gemma family. GaMS continual pre-training may have eroded German relative to base Gemma; verify before relying on it
Gemma 3 12B (vision)	Stamp and seal description	Gemma 3 is multimodal. Useful for producing the descriptive annotations that certified translations require for stamps, seals, and signatures

Licensing differs across these and matters for a legal engagement. GaMS3 inherits the Gemma Terms of Use from its base model, which permit commercial use but impose use restrictions; the Qwen models are Apache 2.0. If licensing must be defensible on the record, review both texts rather than assuming equivalence.

3. Pipeline architecture

A single model invocation per document will not produce usable output for this workload. The following five stages should be built and validated independently, in order. Each stage has a distinct failure mode and a distinct verification step.

Stage	Function	Principal risk
1	Text acquisition (OCR)	Silent character errors in names, dates, case numbers, amounts
2	Segmentation and project setup	Loss of formatting and layout on reassembly
3	Terminology capture	Inconsistent rendering of the same legal term across documents
4	Machine translation	Fluent but unfaithful output; omission without warning
5	Reassembly and quality assurance	Numeric and named-entity drift between source and target

4. Stage 1 — OCR is the dominant risk

This deserves more engineering attention than the translation stage. The reason is a specific and under-appreciated interaction: a language model presented with corrupted OCR output will not flag the corruption. It will silently produce a fluent, plausible translation of the corrupted text. A misread digit in a date, an account number, or a monetary amount propagates into the target document looking entirely correct.

Consequence. The OCR text layer must be treated as a verified artifact in its own right, checked by a human against the page images before translation begins. It cannot be treated as an intermediate that only the machine reads.

Slovene diacritics compound this. The characters č, š, and ž are routinely mangled by OCR engines configured for English, and the resulting words may still be valid Slovene, so no spell-check will catch them.

Recommended stack, all offline:

```
sudo apt install ocrmypdf tesseract-ocr-slv tesseract-ocr-deu tesseract-ocr-eng

# OCR while preserving the original page image underneath
ocrmypdf -l slv+eng --rotate-pages --deskew --clean \
    --output-type pdfa input.pdf output.pdf

# extract the text layer for verification
pdftotext -layout output.pdf output.txt
```

Two practices are worth adopting regardless of tooling. First, keep the page image alongside every translated segment so the translator can check against the original without leaving the tool. Second, extract all numbers, dates, and proper nouns into a separate checklist for targeted verification, since these carry the highest consequence and the lowest redundancy — context will not reveal an error in them.

If Tesseract output proves inadequate on poor scans, evaluate a modern layout-aware alternative such as Surya before accepting the quality loss. This should be tested on a representative sample of the worst scans in the corpus, not the best.

5. Stage 2 — Use a CAT tool, not bespoke scripts

The instinct to build DOCX and XLSX round-trip handling directly is understandable but is not the efficient path. Computer-assisted translation tools already solve format-preserving extraction and reinsertion, and they solve it better than a purpose-built script will, because formatting inside a DOCX paragraph is fragmented across runs in ways that are tedious to reconstruct correctly.

OmegaT is the appropriate choice: free, open source, fully offline, cross-platform, and it handles DOCX and XLSX natively with formatting preserved. It provides translation memory, glossary enforcement, and segment-level review, which is precisely the shape of a machine-translation post-editing workflow.

The decisive argument is not technical, however. The certifying translator will need to review, edit, and attest this work. Delivering it inside a standard CAT project rather than as loose files means they can work in a familiar environment, the translation memory accumulates across all one hundred documents, and their corrections propagate to every later occurrence of the same segment. A bespoke pipeline delivers none of that.

Integration caveat. The mechanism for feeding a local model into OmegaT should be verified before committing. The robust and tool-agnostic approach is to pre-translate segments externally and import the result as a TMX translation memory, which requires no plugin and works with any CAT tool. Direct plugin integration may also exist but was not verified for this assessment.

6. Stage 3 — Terminology is the main quality lever

For legal translation, consistency of terminology matters more than sentence-level elegance, and it is the dimension on which machine output most reliably disappoints. The same Slovene term rendered three different ways across three documents creates ambiguity that a court may treat as substantive.

Build the glossary before bulk translation, not after:

- Extract candidate terms from the corpus — charges, statutory references, court and agency names, procedural terms, party designations, document types.

- Have the certifying translator fix the target rendering for each, once, up front. This is a short task that pays back across every document.
- Load the glossary into the CAT tool for enforcement, and inject the relevant subset into the model prompt for each segment so the model produces the agreed term rather than a synonym.
- Leave institutional names, statutory citations, and case numbers untranslated by policy, with the convention agreed in advance. These are the segments most likely to be silently and confidently mistranslated.

7. Stage 4 — The spreadsheet

A table of roughly 7,000 rows with up to five text columns implies on the order of 35,000 cells. Translated naively at the throughput estimated in Section 2.2, this would run for days.

It almost certainly should not be translated naively. Tabular data of this kind is typically dominated by repeated categorical values — statuses, document types, locations, party names, offence codes. Deduplication before translation is likely to reduce the workload by an order of magnitude, and it delivers guaranteed internal consistency as a side effect, because each distinct string is translated exactly once.

Measure the redundancy before designing anything further:

```
python3 - <<'EOF'
import openpyxl, collections
wb = openpyxl.load_workbook("source.xlsx", read_only=True)
ws = wb.active
cells = [str(c.value).strip() for row in ws.iter_rows()
         for c in row if c.value and not str(c.value).replace(".", "").isdigit()]
u = collections.Counter(cells)
print(f"text cells: {len(cells):,}    unique: {len(u):,}    ratio:
{len(u)/len(cells):.1%}")
for s, n in u.most_common(15): print(f"{n:6,}    {s[:70]}")
EOF
```

If the unique ratio is below roughly 20 percent, build the translation around a string map: extract unique strings, translate each once, write results back by lookup. Output remains in XLSX, which is the correct decision — ten columns will not transfer usefully into Word.

Free-text columns, if any, will need genuine segment-by-segment treatment and should be identified and separated early, since they will dominate the remaining cost.

8. Confidentiality hardening

The stated requirement is that the documents do not leave the premises. The model choice satisfies this trivially, since local inference makes no network calls. Two residual exposures are worth closing, because they are more likely to cause a breach than the inference stack is.

Full-disk encryption

Case material on a laptop is exposed by theft, not by the network. If the installation is not encrypted, that gap is larger than anything else in this document. Verify:

```
lsblk -o NAME,FSTYPE,MOUNTPOINT | grep -i crypt
sudo cryptsetup status /dev/mapper/* 2>/dev/null
```

Retrofitting encryption to an existing installation is disruptive. If it is absent, an encrypted container for the case material alone is the pragmatic alternative.

Ollama cloud model routing

Recent Ollama versions can route requests to hosted models as well as local ones. A tag pulled or invoked carelessly could transmit document text off the machine. Confirm that every model in use is local, and consider blocking outbound network access for the Ollama process entirely as a structural guarantee rather than a procedural one.

```
ollama list          # confirm no cloud-tagged entries
ollama ps           # confirm what is actually loaded
systemctl show ollama -p Environment
```

A second structural measure worth considering: run the entire translation pipeline with networking disabled at the container or namespace level, so that the confidentiality guarantee does not depend on configuration remaining correct.

9. Questions outside the technical scope

These should be settled before significant engineering effort is invested, because an unfavorable answer to the first would render the rest moot.

- Whether machine-translation pre-translation is permissible for court-certified translations in the relevant jurisdictions, and whether its use must be disclosed. Certification regimes vary, and some require the certifying translator to attest to personal translation rather than review. This is a question for the certifying translator and possibly for the court, not one to resolve by assumption.
- Whether the certifying translator is willing to work from machine-translated drafts at all. Post-editing poor output is slower than translating from scratch, and some professionals decline the arrangement on those grounds. Their agreement determines whether the project has value.
- Whether any evidentiary or disclosure obligation attaches to the intermediate artifacts — OCR text layers, translation memories, model outputs — in a pending prosecution. If so, they should be retained and version-controlled deliberately rather than discarded as scratch.
- Whether handling of the material on this machine is consistent with any protective order or undertaking already in force in the proceedings.

10. Recommended sequence

Do not build the full pipeline first. Validate the assumption that machine output is faster to edit than translating from scratch, on a small sample, before committing.

Phase	Activity	Exit criterion
0	Resolve Section 9 questions	Certifying translator confirms the approach is permissible and acceptable
1	Install GaMS3-12B; translate three representative pages by hand-fed text	Translator judges output materially faster to edit than translating fresh
2	Build and verify the OCR stage on the worst scans in the corpus	Text layer accurate on names, dates, and numbers without correction
3	Extract glossary; agree target renderings	Glossary fixed and loaded
4	Stand up OmegaT project; establish TMX pre-translation path	One full document round-trips with formatting intact

Phase	Activity	Exit criterion
5	Measure spreadsheet redundancy; build string map	Unique-string count known; approach chosen
6	Batch translation	Corpus pre-translated; hand off for certification

Phase 1 is the decision point. If the translator does not find the output helpful at that scale, the correct response is to stop, not to build more infrastructure around it.

11. Commands — immediate next steps

Acquire the primary model. It is not in the Ollama library and must be pulled from Hugging Face as a GGUF:

```
# Q6_K, recommended quantization (~9.7 GB)
ollama pull hf.co/mradermacher/GaMS3-12B-Instruct-GGUF:Q6_K

# verify what arrived
ollama show hf.co/mradermacher/GaMS3-12B-Instruct-GGUF:Q6_K
```

Note that the repository publishes both static and imatrix-weighted quantizations; either is acceptable, and exact tag syntax should be confirmed against the model page at pull time.

Take a baseline on real material, timing both prefill and generation:

```
ollama run hf.co/mradermacher/GaMS3-12B-Instruct-GGUF:Q6_K --verbose

# suggested system framing for legal source text:
# "Translate the following Slovene legal text into English. Preserve
# paragraph structure. Do not translate case numbers, statutory
# citations, or institution names; reproduce them verbatim. Mark any
# illegible or uncertain passage as [ILLEGIBLE]. Output only the
# translation."
```

The instruction to mark uncertainty is not decorative. It is the only lever available against the silent-hallucination failure mode described in Section 4, and its effectiveness should be tested deliberately by feeding the model a deliberately corrupted passage.